INCH-POUND

MIL-PRF-28876/29F
w/AMENDMENT 2
5 March 2026

SUPERSEDING
MIL-PRF-28876/29F(SH)
w/INT. AMENDMENT 1
12 October 2023
MIL-PRF-28876/29F
1 March 2021

PERFORMANCE SPECIFICATION SHEET

CONNECTORS, FIBER OPTIC, CIRCULAR, PLUG AND RECEPTACLE STYLE, MULTIPLE REMOVABLE TERMINI, SCREW THREADS, 90-DEGREE BACKSHELL, ENVIRONMENT RESISTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-28876.

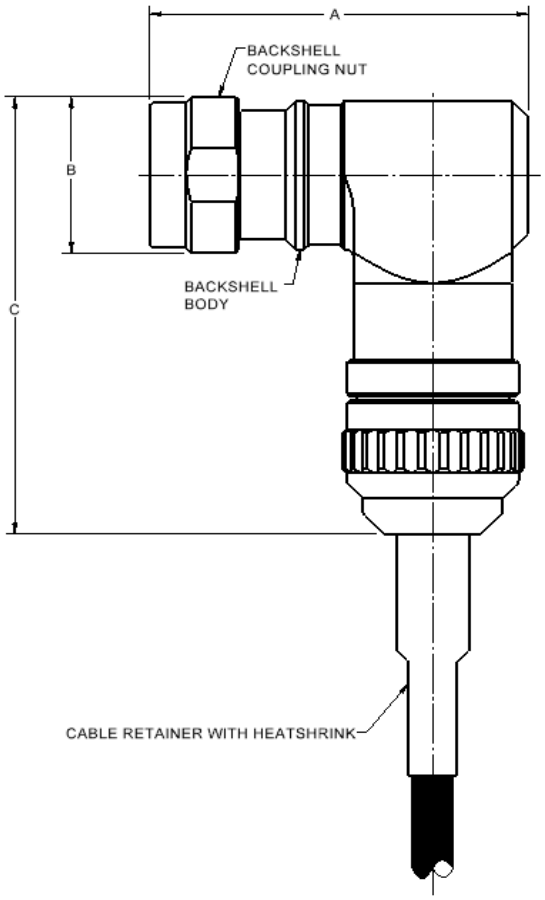


FIGURE 1. 90-degree backshell.

AMSC N/A FSC 6060

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Shell size	Shell size designator	Backshell designator	A max	B max [1/]	C max
11	A	1L, 2L, 3L	4.2 (107)	1.2 (30.5)	3.0 (76)
		1, 2, 3	3.2 (81)	1.2 (30.5)	
13	B	1L, 2L	4.3 (109)	1.2 (30.5)	3.1 (79)
		1, 2	3.2 (81)	1.2 (30.5)	
15	C	1L, 2L, 3L	4.3 (109)	1.5 (38.1)	3.1 (79)
		1, 2, 3	3.4 (86.36)	1.5 (38.1)	
23	F	1, 2, 3	3.9 (99.1)	2.250 (57.15)	4.200 (106.68)

FOOTNOTE:

[1/] Dimension B is maximum for backshell diameter and envelope. See MIL-PRF-28876/9 for M DIA dimensioning.

NOTES:

1. Dimensions are in inches.
2. Metric equivalents (mm) are given in parentheses for general information only.
3. Dimensions apply to plated/finished part.
4. Backshell internal configuration not shown. The backshell interface dimensions shall be in accordance with figure A-7 of MIL-PRF-28876.

FIGURE 1. 90-degree backshell - Continued.

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REQUIREMENTS:

Dimensions and spline configurations. See figure 1 herein and figure A-7 of MIL-PRF-28876. When mated to a plug or receptacle, the backshell shall hold the plug or receptacle insert in proper position (see figures A-1, A-2, and A-6 of MIL-PRF-28876).

Weight. 6.4 ounces (181.44 grams), maximum.

Part or identifying number (PIN). Marked on the coupling ring of the backshell (see table I).

M	**28876**	**/29**	**X**	**X**	**X**
Prefix for Military Specification	Specification Number	Spec Sheet No.	Shell size designator (see table I)	Backshell designator (see table I)	Class designator (see table I)

TABLE I. PIN designators.

Shell size	Shell size designator	Backshell designator	Cable diameter range inch (mm)	Class designator (see table II)
11	A	1L, 1	0.240 – 0.250 (6.10 – 6.35)	T or blank
		2L, 2	0.280 – 0.315 (7.11 – 8.00)	T or blank
		3L, 3	0.305 – 0.346 (7.75 – 8.79)	T or blank
13	B	1L, 1	0.280 – 0.315 (7.11 – 8.00)	T or blank
		2L, 2	0.305 – 0.346 (7.75 – 8.79)	T or blank
15	C	1L, 1	0.495 – 0.515 (12.57 – 13.08)	T or blank
		2L, 2	0.380 – 0.423 (9.65 – 10.74)	T or blank
		3L, 3	0.423 – 0.465 (10.74 – 11.81)	T or blank
		4L, 4	0.305 – 0.346 (7.75 – 8.78)	T or blank
23	F	1	0.768 – 0.866 (19.50 – 22.00)	T or blank
		2	0.900 – 1.000 (22.86 – 25.40)	T or blank

TABLE II. Class designators.

Class designator	Description
Blank	Aluminum with non-conductive finish
T	Aluminum with conductive plating

Example: M28876/29B1

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Fiber optic cable diameter. See table I.

Mating counterpart. Backshell mates with MIL-PRF-28876/1 receptacle, MIL-PRF-28876/6 plug, and MIL-PRF-28876/11 receptacle.

Installation and removal tools. Adjustable or 1.25-inch (31.8-millimeter) open-end wrench, strap wrench, and scissors.

NOTES:

Referenced publications. In addition to MIL-PRF-28876, this specification sheet references the following publications:

MIL-PRF-28876/1 MIL-PRF-28876/6 MIL-PRF-28876/9

MIL-PRF-28876/11

AMENDMENT NOTATIONS. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
DLA – CC

Preparing activity:
Navy – SH
(Project 6060-2023-030)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.

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